Chapter 1

Clinical Significance of Dental Anatomy, Histology, Physiology and Occlusion

"Success in life is founded upon attention to the smallest of things… rather than to the largest of things…"

—Booker T. Washington

A thorough understanding of the histology, physiology and occlusal interactions of the dentition and supporting tissues is essential for the restorative dentist. Knowledge of the structures of teeth (enamel, dentin, cementum and pulp) and their relationships to each other and to the supporting structures is necessary, especially when treating dental caries. Proper tooth form contributes to healthy supporting tissues. The relationships of form to function are especially noteworthy when considering the shape of the dental arch, proximal contacts, occlusal contacts and mandibular movement.

Teeth and Supporting Tissues

Dentitions

Humans have *primary* and *permanent* dentitions. The primary dentition consists of 10 maxillary and 10 mandibular teeth. Primary teeth exfoliate and are replaced by the permanent dentition, which consists of 16 maxillary and 16 mandibular teeth.

Classes of Human Teeth: Form and Function

Human teeth are divided into classes on the basis of form and function. The primary and permanent dentitions include the *incisor*, *canine* and *molar* classes. The fourth class, the *premolar*, is found only in the permanent dentition (Fig. 1.1). Tooth form predicts the function of teeth; class traits are the characteristics that place teeth into functional categories. Because the diet of humans consists of animal and plant foods, the human dentition is called *omnivorous*.

Incisors

The incisors are located near the entrance of the oral cavity and function as cutting or shearing instruments for food (Fig. 1.1). From a proximal view, the crowns of these teeth have a relatively triangular shape, with a narrow incisal surface and a broad cervical base. During mastication, incisors are used to shear (cut through) food.

Clinical Notes

Incisors are essential for the proper esthetics of the smile, facial soft tissue contours (e.g. lip support) and speech (phonetics).

Canines

Canines possess the longest roots of all teeth and are located at the corners of the dental arch. They function in the seizing, piercing, tearing and cutting of food. From a proximal view, the crown also has a triangular shape, with a thick incisal ridge. The anatomic form of the crown and the length of the root make these teeth strong, stable abutment teeth for a fixed or removable prosthesis.

Clinical Notes

Canines not only serve as important guides in occlusion because of their anchorage and position in the dental arches but also play a crucial role (along with the incisors) in the esthetics of smile and lip support (Fig. 1.1).

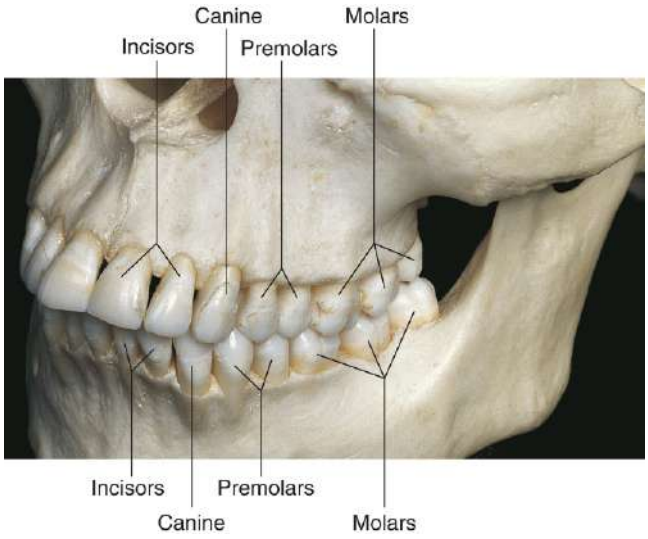


Figure 1.1 Maxillary and mandibular teeth in maximum inter-cuspal position. The classes of teeth are incisors, canines, premolars and molars. Cusps of mandibular teeth are one-half cusp anterior of corresponding cusps of teeth in the maxillary arch.
(From Logan BM, Reynolds P, Hutchings RT: *McMinn's color atlas of head and neck anatomy,* ed 4, Edinburgh, 2010, Mosby.)

Premolars

Premolars serve a dual role:

I. They are similar to canines in the tearing of food.
II. They are similar to molars in the grinding of food.

The occlusal surfaces of the premolars present a series of curves in the form of concavities and convexities that should be maintained throughout life for correct occlusal contacts and function.

Clinical Notes

Although less visible than incisors and canines, premolars still can play an important role in esthetics.

Molars

Molars are large, multi-cusped, strongly anchored teeth located nearest to the temporomandibular joint (TMJ), which serves as the fulcrum during function. These teeth have a major role in the crushing, grinding and chewing of food to the smallest dimensions suitable for swallowing. They are well suited for this task because they have broad occlusal surfaces and multi-rooted anchorage (Fig. 1.2).

Clinical Notes

Premolars and molars are important in maintaining the vertical dimension of the face (Fig. 1.1).

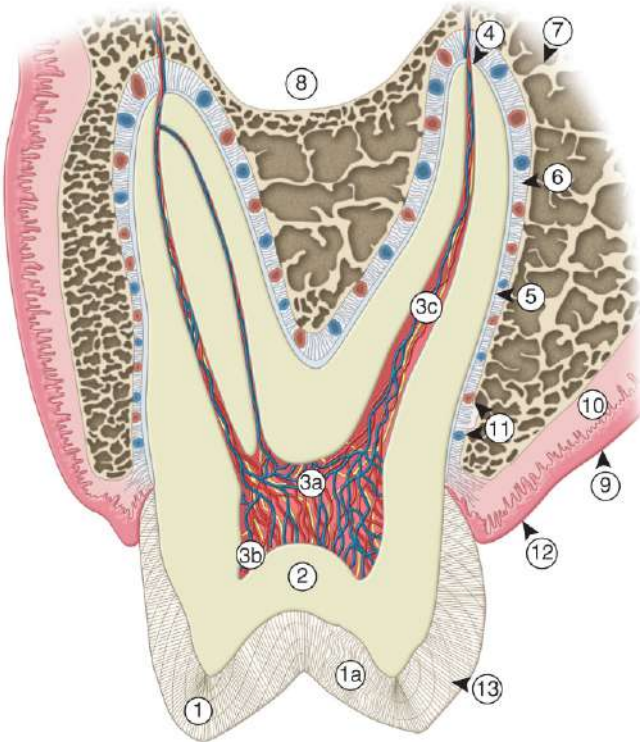


Figure 1.2 Cross-section of the maxillary molar and its supporting structures. *1*, Enamel; *1a*, gnarled enamel; *2*, dentin; *3a*, pulp chamber; *3b*, pulp horn; *3c*, pulp canal; *4*, apical foramen; *5*, cementum; *6*, periodontal fibres in periodontal ligament; *7*, alveolar bone; *8*, maxillary sinus; *9*, mucosa; *10*, submucosa; *11*, blood vessels; *12*, gingival and *13*, striae of Retzius.

Structures of Teeth

Teeth are composed of enamel, the pulp–dentin complex and cementum (Fig. 1.2). Each of these structures is discussed individually.

Enamel

Enamel formation, *amelogenesis*, is accomplished by cells called *ameloblasts*. These cells originate from the embryonic germ layer known as *ectoderm*. Enamel covers the anatomic crown of the tooth and varies in thickness in different areas (Fig. 1.2). It is thicker at the incisal and occlusal areas of a tooth and becomes progressively thinner until it terminates at the cementoenamel junction (CEJ). The thickness also varies from one class of tooth to another:

- Averaging 2 mm at the incisal ridges of incisors.
- 2.3–2.5 mm at the cusps of premolars.
- 2.5–3 mm at the cusps of molars.

The cusps of posterior teeth begin as separate ossification centres, which form lobes that coalesce. Enamel usually decreases in thickness towards the junction of these developmental features and can approach zero where the junction is fissured (non-coalesced).

Chemical composition

Enamel is a highly mineralized crystalline structure.

- *Hydroxyapatite*, in the form of a crystalline lattice, is the largest mineral constituent (90%–92% by volume).
- Organic matrix proteins (1%–2% by volume).
- Water (4%–12% by volume).

Structure of enamel

Enamel is composed of millions of *enamel rods* or *prisms*, which are the largest structural components, *rod sheaths* and a cementing inter-rod substance in some areas.

Human enamel is composed of rods that, in transverse section, have a rounded head or *body section* and a *tail section*, forming a repetitive series of interlocking prisms. The rounded head portion of each prism (5 µm wide) lies between the narrow tail portions (5 µm long) of two adjacent prisms (Fig. 1.3). Generally, the rounded head portion is oriented in the incisal or occlusal direction; the tail section is oriented cervically.

The rods vary in number from approximately 5 million for a mandibular incisor to about 12 million for a maxillary molar. The rods are densely packed and intertwined in a wavy course, and each extends from the dentinoenamel junction (DEJ) to the external surface of the tooth. Enamel rod diameter near the dentinal borders is about 4 µm and about 8 µm near the surface. This difference accommodates the larger outer surface of the enamel crown compared with the dentinal surface at the DEJ.

Clinical Notes

- Enamel is the *hardest substance* of the human body.
- Enamel rods that lack dentin support because of caries or improper preparation design are easily fractured away from neighbouring rods. For optimal strength in tooth preparation, all enamel rods should be supported by dentin (Fig. 1.4).

Apatite crystallites

The structural components of the enamel prism are millions of small, elongated *apatite crystallites* that vary in size and shape. The crystallites are tightly packed in a distinct pattern of orientation that gives strength and structural identity to the enamel prisms. The crystallites are irregular in shape, with an average length of 160 nm and an average width of 20–40 nm. Each apatite crystallite is composed of thousands of *unit cells* that have a highly ordered arrangement of atoms. A crystallite may be 300 unit cells long, 40 cells wide and 20 cells thick in a hexagonal configuration. An organic matrix or prism sheath also surrounds individual crystals and appears to be an organically rich interspace rather than a structural entity.

Gnarled enamel

Enamel rods follow a wavy, spiralling course, producing an alternating arrangement for each group or layer of rods as they change direction in progressing from the dentin to the enamel surface, where they end a few micrometres short of the tooth surface. They initially follow a curving path through one-third of the enamel next to the DEJ. After that, the rods usually follow a more direct path through the remaining two-thirds of the enamel to the enamel surface. Groups of enamel rods may entwine with adjacent

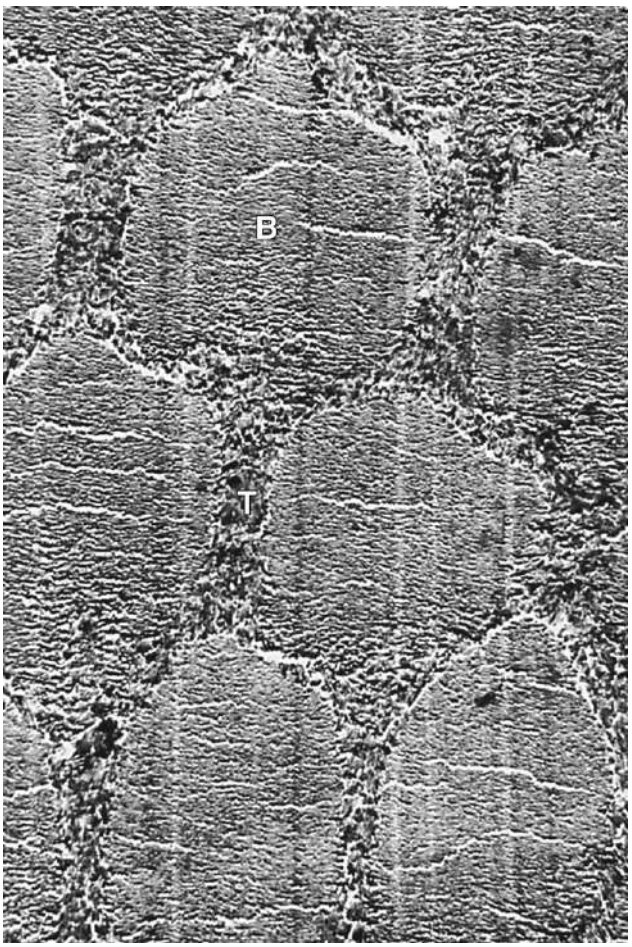


Figure 1.3 Electron micrograph of cross-section of rods in mature human enamel. Crystal orientation is different in 'bodies' (*B*) than in 'tails' (*T*). Approximate level of magnification 5000×.
(From Meckel AH, Griebstein WJ, Neal RJ: Structure of mature human dental enamel as observed by electron microscopy, *Arch Oral Biol* 10(5):775–783, 1965.)

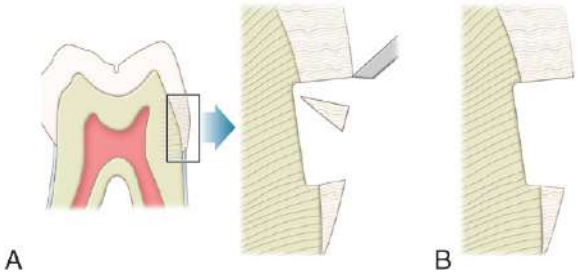


Figure 1.4 (A) Enamel rods unsupported by dentin are fractured away readily by pressure from hand instrument. (B) Cervical preparation showing enamel rods supported by dentin.



Figure 1.5 Gnarled enamel.
(From Berkovitz BKB, Holland GR, Moxham BJ: *Oral anatomy, histology and embryology*, ed 4, Edinburgh, 2009, Mosby.)

groups of rods, and they follow a curving irregular path towards the tooth surface. These constitute *gnarled enamel*, which occurs near the cervical regions and the incisal and occlusal areas (Fig. 1.5). Gnarled enamel is not subject to fracture as much as a regular enamel. This type of enamel formation does not yield readily to the pressure of bladed, hand-cutting instruments in tooth preparation.

Clinical Notes

The orientation of the enamel rod heads and tails and the gnarling of enamel rods provide strength by resisting, distributing and dissipating impact forces.

Hunter–Schreger bands

The changes in the direction of enamel prisms that minimize fracture in the axial direction produce an *optical appearance* called *Hunter–Schreger bands* (Fig. 1.6). These bands appear to be composed of alternate light and dark zones of varying widths that have slightly different permeability and organic content. These bands are found in different areas of each class of teeth. Because the enamel rod orientation varies in each tooth, Hunter–Schreger bands also have a variation in the number present in each tooth. In anterior teeth, they are located near the incisal surfaces. In molars, the bands occur from near the cervical region to the cusp tips.

Enamel tufts

Enamel tufts are *hypomineralized* structures of the enamel rods and the inter-rod substance that project between adjacent groups of enamel rods from the DEJ (Fig. 1.7). These projections arise in dentin, extend into enamel in the direction of the long axis of the crown and may play a role in the spread of dental caries.

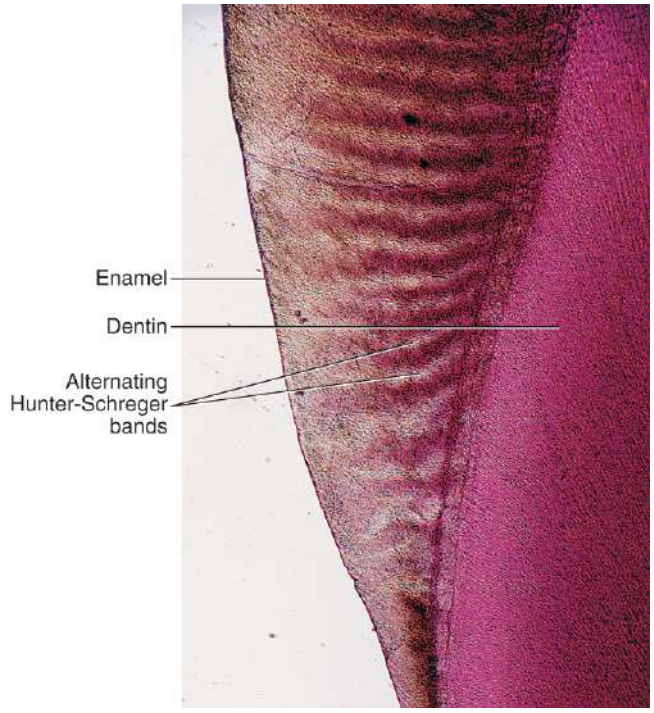


Figure 1.6 Photomicrograph of enamel photographed by reflected light of Hunter–Schreger bands.
(From Avery JK, Chiego DJ: *Essentials of oral histology and embryology: a clinical approach*, ed 3, St Louis, MO, 2006, Mosby.)

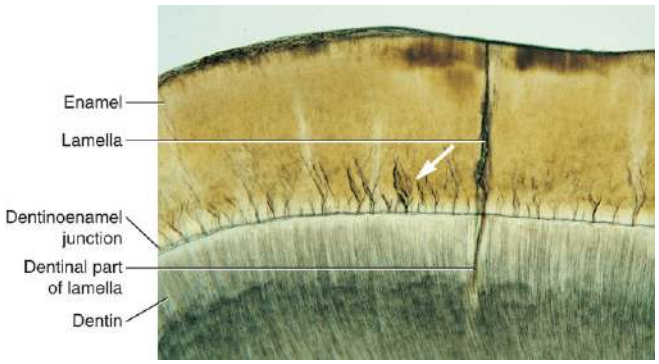


Figure 1.7 Microscopic view through lamella that goes from enamel surface into dentin. Note the enamel tufts *(arrow)*.
(From Bath Balogh M, Fehrenbach MJ: *Illustrated dental embryology, histology, and anatomy*, ed 3, St Louis, MO, 2011, Saunders. Courtesy: James McIntosh, PhD, Assistant Professor Emeritus, Department of Biomedical Sciences, Baylor College of Dentistry, Dallas, TX.)

Enamel lamellae

Enamel lamellae are *thin, leaf-like faults* between enamel rod groups that extend from the enamel surface to the DEJ, sometimes extending into dentin (Fig. 1.7). They contain mostly organic material, which is a weak area predisposing a tooth to the entry of bacteria and dental caries.

Striae of Retzius

Enamel rods are formed linearly by successive apposition of enamel in discrete increments. The resulting variations in structure and mineralization are called *incremental*

striae of Retzius and can be considered *growth rings* (Fig. 1.2). In horizontal sections of a tooth, the striae of Retzius appear as concentric circles. In vertical sections, the lines traverse the cuspal and incisal areas in a symmetric arc pattern, descending obliquely to the cervical region and terminating at the DEJ. When these circles are incomplete at the enamel surface, a series of alternating grooves, called *imbrication lines of Pickerill*, are formed. The elevations between the grooves are called *perikymata*; these are continuous around a tooth and usually lie parallel to the CEJ and each other.

Nasmyth's membrane or primary enamel cuticle

Once damaged, enamel is incapable of repairing itself because the ameloblast cell degenerates after the formation of the enamel rod. The *final act* of the ameloblast is secretion of a membrane covering the end of the enamel rod. This layer is referred to as *Nasmyth's membrane* or *primary enamel cuticle*. This membrane covers the newly erupted tooth and is worn away by mastication and cleaning. This membrane is replaced by an organic deposit called the *pellicle*, which is a precipitate of salivary proteins. Microorganisms may attach to the pellicle to form bacterial plaque, which, if acidogenic in nature, can be a potential precursor to dental disease.

Prismless enamel

A structureless outer layer of enamel about 30 µm thick is found most commonly towards the cervical area and less often on cusp tips. No prism outlines are visible, and all of the apatite crystals are parallel to one another and perpendicular to the striae of Retzius. This layer, referred to as *prismless enamel*, may be more heavily mineralized. Microscopically, the enamel surface initially has circular depressions indicating where the enamel rods end. These concavities vary in depth and shape, and they may contribute to the adherence of plaque material, with a resultant caries attack, especially in young individuals. The dimpled surface anatomy of the enamel, however, gradually wears smooth with age.

Dentinoenamel junction

The interface of enamel and dentin is known as dentinoenamel junction or DEJ. It is scalloped or wavy in outline, with the crest of the waves penetrating towards enamel (Fig. 1.8). The rounded projections of enamel fit into the shallow depressions of dentin. This interdigitation may contribute to the firm attachment between dentin and enamel.

Enamel is the hardest substance of the human body. Hardness may vary over the external tooth surface according to the location; also, it decreases inward, with hardness lowest at the DEJ. The density of enamel also decreases from the surface to the DEJ. Enamel is a rigid structure that is both strong and brittle (high elastic modulus, high compressive strength and low tensile strength). The ability of the enamel to withstand masticatory forces

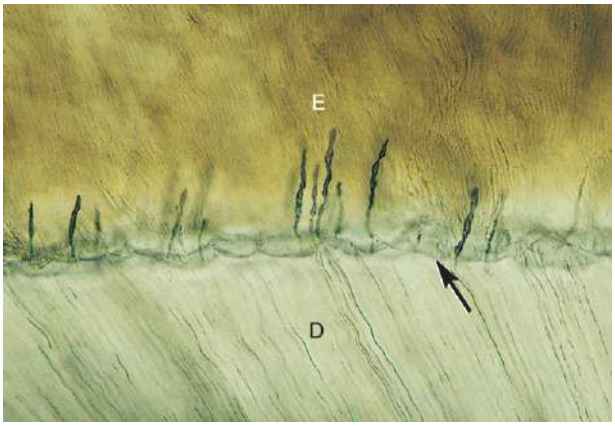


Figure 1.8 Microscopic view of scalloped dentoenamel junction (DEJ) *(arrow)*. *E*, Enamel; *D*, dentin.
(From Bath Balogh M, Fehrenbach MJ: *Illustrated dental embryology, histology, and anatomy*, ed 3, St. Louis, MO, 2011, Saunders. Courtesy: James McIntosh, PhD, Assistant Professor Emeritus, Department of Biomedical Sciences, Baylor College of Dentistry, Dallas, TX.)

depends on a stable attachment to the dentin by means of the DEJ. Dentin is a more flexible substance that is strong and resilient (low elastic modulus, high compressive strength and high tensile strength), which essentially increases the fracture toughness of the more superficial enamel.

The DEJ is approximately 2 µm wide and comprises a mineralized complex of interwoven dentin and enamel matrix proteins. In addition to the physical, scalloped relationship between the enamel and dentin, an interphase matrix layer (made primarily of a fibrillary collagen network) extends 100–400 µm from the DEJ to the enamel. This matrix-modified interphase layer is considered to provide fracture propagation limiting properties to the interface between the enamel and the DEJ and thus overall structural stability of the enamel attachment to dentin.

Pit and fissure

The occlusal surfaces of premolars and molars have grooves and fossae that form at the junction of the developmental lobes of enamel. Failure of the enamel of the developmental lobes to coalesce results in a deep invagination of the enamel surface and is termed *fissure*. Non-coalesced enamel at the deepest point of a fossa is termed as *pit*. These pit and fissures act as food and bacterial traps that predispose the tooth to dental caries (Fig. 1.9).

Enamel maturation

Although enamel is a hard, dense structure, it is permeable to certain ions and molecules. The route of passage may be through structural units that are hypomineralized and rich in organic content, such as rod sheaths, enamel cracks and other defects. Water plays an important role as a transporting medium through small inter-crystalline spaces. Enamel permeability decreases with age because of changes in the enamel matrix, a decrease referred to as *enamel maturation*.

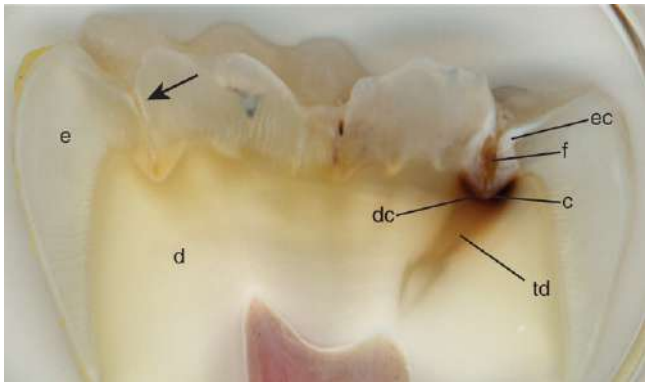


Figure 1.9 Fissure (*f*) at junction of lobes allows accumulation of food and bacteria predisposing the tooth to dental caries (*c*); enamel (*e*); dentin (*d*); enamel caries lesion (*ec*); dentin caries lesion (*dc*); transparent dentin (*td*); early enamel demineralization *(arrow).*

Clinical Notes

Effect of fluoride on enamel: Enamel is soluble when exposed to acidic conditions, but the dissolution is not uniform. Solubility of enamel increases from the enamel surface to the DEJ. When fluoride ions are present during enamel formation or are topically applied to the enamel surface, the solubility of surface enamel is decreased. Fluoride can affect the chemical and physical properties of the apatite mineral and influence the hardness, chemical reactivity and stability of enamel while preserving the apatite structures. Trace amounts of fluoride stabilize enamel by

- lowering acid solubility;
- decreasing the rate of demineralization and
- enhancing the rate of remineralization.

Translucency of enamel

Because enamel is semitranslucent, the colour of a tooth depends on

- colour of the underlying dentin;
- thickness of the enamel and
- amount of stain in the enamel.

The amount of translucency of enamel is related to variations in the degree of calcification and homogeneity. Abnormal conditions of enamel usually result in aberrant colour. Enamel becomes temporarily whiter within minutes when a tooth is isolated from the moist oral environment by a rubber dam or cotton rolls. This change in colour is explained by the temporary loss of loosely bound (or exchangeable) water (<1% by weight).

Clinical Notes

The shade of a tooth must be determined *before* isolation and preparation of a tooth for a tooth-coloured restoration.

Pulp–Dentin Complex

Dentin and pulp tissues are specialized connective tissues of *mesodermal origin*, formed from the dental papilla of the tooth bud. Many investigators consider these two tissues as a single tissue, which form the pulp–dentin complex, with mineralized dentin constituting the mature end product of cell differentiation and maturation.

Pulp

The pulp is circumscribed by the dentin and is lined peripherally by a cellular layer of odontoblasts adjacent to dentin. Anatomically, the pulp is divided into the following:

I. ***Coronal pulp*** located in the pulp chamber in the crown portion of the tooth, including the pulp horns that are directed towards the incisal ridges and cusp tips.
II. ***Radicular pulp*** located in the pulp canals in the root portion of the tooth. The radicular pulp is continuous with the periapical tissues by connecting through the apical foramen or foramina of the root. Accessory canals may extend from the pulp canals laterally through the root dentin to the periodontal tissues. The shape of each pulp conforms generally to the shape of each tooth (Fig. 1.2).

The pulp contains nerves, arterioles, venules, capillaries, lymph channels, connective tissue cells, intercellular substance, odontoblasts, fibroblasts, macrophages, collagen and fine fibres. The pulp is circumscribed peripherally by a specialized odontogenic area composed of the odontoblasts, the cell-free zone and the cell-rich zone.

Functions of pulp

The dental pulp occupies the pulp cavity in the tooth and is a unique, specialized organ of the human body that serves four functions:

I. ***Formative or developmental***. The formative function is the production of primary and secondary dentin by odontoblasts.
II. ***Nutritive***. The nutritive function supplies nutrients and moisture to dentin through the blood vascular supply to the odontoblasts and their processes.
III. ***Sensory or protective.*** The sensory function provides nerve fibres within the pulp to mediate the sensation of pain. Dentin receptors are unique because various stimuli elicit only pain as a response. The pulp usually does not differentiate between heat, touch, pressure and chemicals. Motor fibres initiate reflexes in the muscles of the blood vessel walls for the control of circulation in the pulp.
IV. ***Defensive or reparative.*** The defensive function of the pulp is related primarily to its response to irritation by mechanical, thermal, chemical or bacterial stimuli. The deposition of reparative dentin acts as a protective barrier against caries and various other irritating factors. In cases of severe irritation, the pulp responds by an

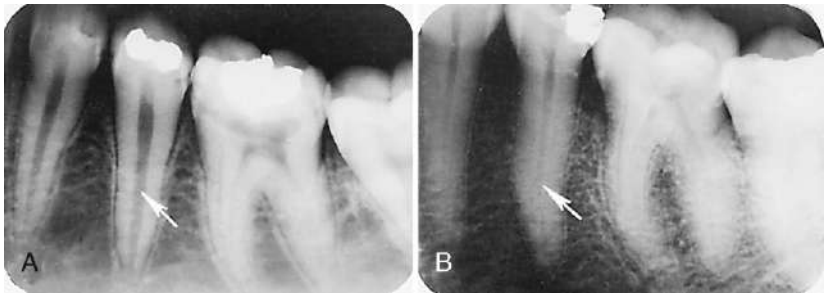


Figure 1.10 Pulp cavity size. (A) Premolar radiograph of young person. (B) Premolar radiograph of older person. Note the difference in the size of the pulp cavity *(arrows)*.

inflammatory reaction similar to that for any other soft tissue injury. The inflammation may become irreversible, however, and can result in the death of the pulp because the confined, rigid structure of the dentin limits the inflammatory response and the ability of the pulp to recover.

Clinical Notes

Knowledge of the contour and size of the pulp cavity is essential during tooth preparation. In general, the pulp cavity is a miniature contour of the external surface of the tooth. Pulp cavity size varies with tooth size among individuals and even within a single person. With advancing age, the pulp cavity usually decreases in size. Radiographs are an invaluable aid in determining the size of the pulp cavity and any existing pathologic condition (Fig. 1.10). A primary objective during operative procedures must be the preservation of the health of the pulp.

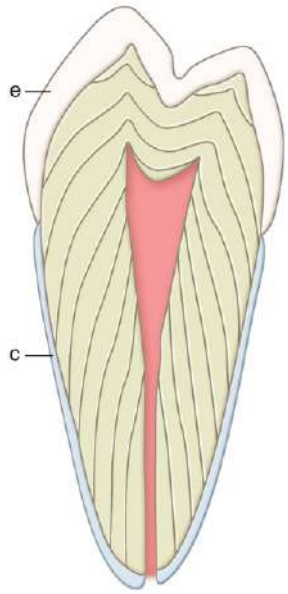


Figure 1.11 Pattern of formation of primary dentin. This figure also shows enamel (*e*) covering the anatomic crown of the tooth and cementum (*c*) covering the anatomic root.

Dentin

Dentin forms the largest portion of the tooth structure, extending almost the full length of the tooth. Externally, dentin is covered by enamel on the anatomic crown and cementum on the anatomic root. Internally, dentin forms the walls of the pulp cavity (pulp chamber and pulp canals) (Fig. 1.11).

Dentinogenesis

Dentin formation or *dentinogenesis* is accomplished by cells called *odontoblasts*. Odontoblasts are considered as a part of pulp and dentin tissues because their cell bodies are in the pulp cavity, but their long, slender cytoplasmic cell processes (Tomes' fibres) extend well (100–200 µm) into the tubules in the mineralized dentin (Fig. 1.12). Because of these odontoblastic cell processes, dentin is considered a *living tissue*, with the capability of reacting to physiologic and pathologic stimuli. Dentin formation begins at areas subjacent to the cusp tip or incisal ridge and gradually spreads to the apex of the root (Fig. 1.11). In contrast to enamel formation, dentin formation continues after tooth eruption and throughout the life of the pulp.

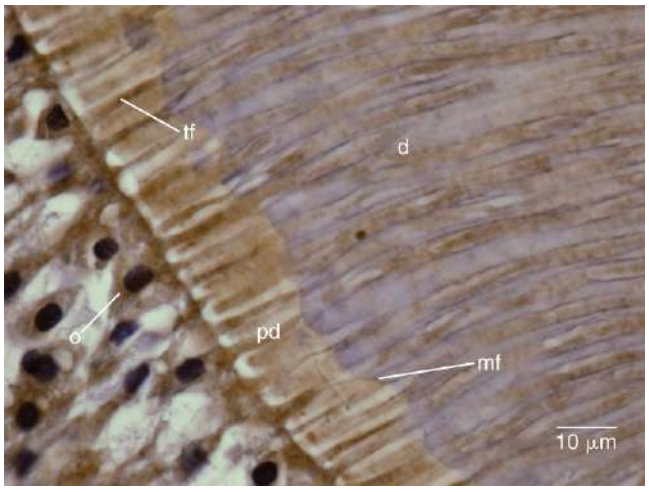


Figure 1.12 Odontoblasts (*o*) have cell processes (Tomes' fibres [tf]) that extend through the predentin (*pd*) into dentin (*d*). *mf*, Mineralization front.

Predentin

Dentin formation begins immediately before enamel formation. The most recently formed layer of dentin is always on the pulpal surface. This unmineralized zone of dentin is immediately next to the cell bodies of odontoblasts and is called *predentin*.

Enamel spindles

Odontoblastic processes occasionally cross the DEJ into enamel; these are termed *enamel spindles* when their ends are thickened (Fig. 1.13). They may serve as pain receptors, explaining the enamel sensitivity experienced by some patients during tooth preparation.

Dentinal tubules

The dentinal tubules are small canals that extend through the entire width of dentin, from the pulp to the DEJ (Fig. 1.14). Each tubule contains the cytoplasmic cell process (Tomes' fibre) of an odontoblast and is lined with a layer of *peritubular dentin*, which is much more mineralized than the surrounding *intertubular dentin* (Fig. 1.14).

The course of the dentinal tubules is a slight 'S'-curve in the tooth crown, but the tubules are straighter in the incisal ridges, cusps and root areas (Fig. 1.15). Near the DEJ, the tubules divide into several terminal branches, forming an intercommunicating and anastomosing network (Fig. 1.16).

Clinical Notes

- The surface area of dentin is much larger at the DEJ or dentinocemental junction than it is on the pulp cavity side. Because odontoblasts form dentin while progressing inward towards the pulp, the tubules are forced closer together. The number of tubules *increase* from 15,000 to 20,000/mm^2 at the DEJ to 45,000 to 65,000/mm^2 at the pulp.[1]
- The lumen of the tubules also varies from the DEJ to the pulp surface. In coronal dentin, the average diameter of tubules at the DEJ is 0.5–0.9 µm, but this increases to 2–3 µm near the pulp (Fig. 1.17).

Secondary dentin

After the primary dentin is formed, dentin deposition continues at a reduced rate even without obvious external stimuli, although the rate and amount of this physiologic *secondary dentin* vary considerably among individuals.

In the secondary dentin, the tubules take a slightly *different directional pattern* in contrast to primary dentin (Fig. 1.18). Secondary dentin forms on all internal aspects of the pulp cavity, but in the pulp chamber, in multi-rooted teeth, it tends to be thicker on the roof and floor than on the side walls.[2]

Tertiary dentin

The pulp–dentin complex responds to tooth pathology through pulpal immune-inflammation defence systems and dentin repair/formation. Deep dentin formation processes occur simultaneously with the pulpal inflammatory response and result in the generation of *tertiary dentin* at the pulp–dentin interface. The net effect of these processes is to increase the thickness/effectiveness of the dentin as a protective barrier for the pulp tissue. Two types of tertiary dentin form in response to lesion formation.

1. *Reactionary tertiary dentin*. In the case of mild injury (e.g. a shallow caries lesion), primary odontoblasts



Figure 1.13 Longitudinal section of enamel. Odontoblastic processes extend into enamel as enamel spindles (*A*).
(From Berkovitz BKB, Holland GR, Moxham BJ: *Oral anatomy, histology and embryology*, ed 4, Edinburgh, 2009, Mosby. Courtesy: Dr. R Sprinz.)

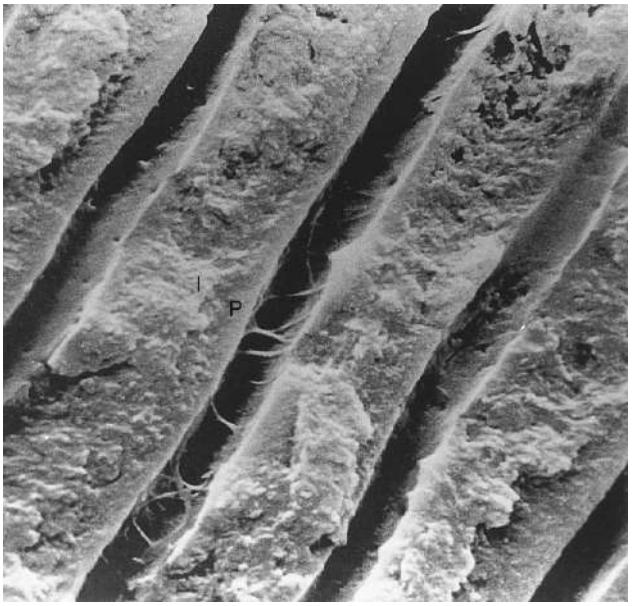


Figure 1.14 Dentinal tubules in cross-section, 1.2 mm from pulp. Peritubular dentin (*P*) is more mineralized than intertubular dentin (*I*).
(From Brännström M: *Dentin and pulp in restorative dentistry*, London, 1982, Wolfe Medical.)



Figure 1.15 Ground section of human incisor. Course of dentinal tubules is in a slight S-curve in the crown, but straight at the incisal tip and in the root.
(From Young B, Lowe JS, Stevens A, Heath JW: *Wheater's functional histology: a text and colour atlas*, ed 5, Edinburgh, 2006, Churchill Livingstone.)

initiate increased formation of dentin along the internal aspect of the dentin beneath the affected area through secretion of *reactionary tertiary dentin* (or reactionary dentin). Reactionary dentin is tubular in nature and is continuous with primary and secondary dentin.

II. *Reparative tertiary dentin*. More severe injury (e.g. a deep caries lesion) causes the death of the primary

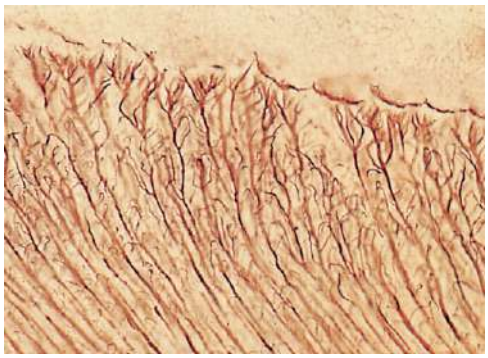


Figure 1.16 Ground section showing dentinal tubules and their lateral branching close to the dentinoenamel junction (DEJ).
(From Berkovitz BKB, Holland GR, and Moxham BJ: *Oral anatomy, histology, and embryology*, ed 4, Edinburgh, 2010, Mosby.)

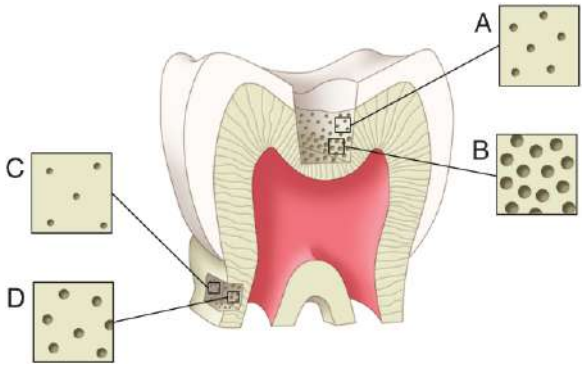


Figure 1.17 Tubules in superficial dentin close to the dentinoenamel junction (DEJ) (A) are smaller and more sparsely distributed compared with deep dentin (B). The tubules in superficial root dentin (C) and deep root dentin (D) are smaller and less numerous than those in comparable depths of coronal dentin.

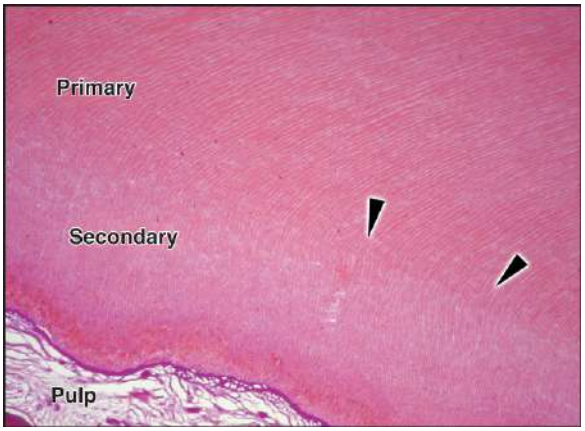


Figure 1.18 Ground section of dentin with pulpal surface at right. Dentinal tubules curve sharply *(arrows)* as they move from primary to secondary dentin. Dentinal tubules are more irregular in shape in secondary dentin.
(From Nanci A: *Ten Cate's oral histology: development, structure, and function*, ed 7, St Louis, MO, 2008, Mosby.)

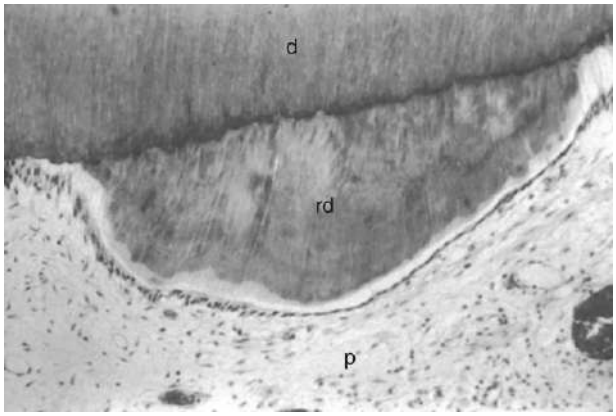


Figure 1.19 Reparative dentin in response to a carious lesion. (From Trowbridge HO: Pulp biology: progress during the past 25 years, *Aust Endo J* 29(1):5–12, 2003.)

odontoblasts. When therapeutic steps successfully resolve the injury, replacement cells (variously referred to as *secondary odontoblasts*, *odontoblast-like cells* or *odontoblastoid cells*) differentiate from pulpal mesenchymal cells. The secondary odontoblasts subsequently generate *reparative tertiary dentin* (or reparative dentin) as a part of the ongoing host defence. Reparative dentin usually appears as a localized dentin deposit on the wall of the pulp cavity immediately subjacent to the area on the tooth that had received the injury (Fig. 1.19).

Sclerotic dentin

Sclerotic dentin results from aging or mild irritation (e.g. slowly advancing caries) and causes a change in the composition of the primary dentin. The peritubular dentin becomes wider, gradually filling the tubules with calcified material, progressing pulpally from the DEJ. These areas are harder, denser, less sensitive and more protective of the pulp against subsequent irritations.

I. Sclerosis resulting from aging is called *physiologic dentin sclerosis*.
II. Sclerosis resulting from a mild irritation is called *reactive dentin sclerosis*. Reactive dentin sclerosis often can be seen radiographically in the form of a more radiopaque (lighter) area in the 'S'-shape of the tubules.

Chemical composition of dentin

Dentin is less mineralized than enamel but more mineralized than cementum or bone. Human dentin is composed of approximately (by weight)

- ***70% inorganic material***. Hydroxyapatite crystallites are arranged in a less systematic manner than enamel crystallites.
- ***20% organic material***. Approximately 90% type I collagen and 10% non-collagenous proteins.
- ***10% water***.

Dentinal crystallites are smaller than enamel crystallites, having a length of 20–100 nm and a width of about 3 nm, which is similar to the size seen in bone and cementum.[2]

Physical properties

I. The hardness of dentin averages one-fifth that of enamel, and its hardness near the DEJ is about 3 times greater than near the pulp.
II. Dentin becomes harder with age, primarily as a result of increase in mineral content.
III. Dentin is a hard, mineralized tissue; it is flexible, with a *modulus of elasticity* of approximately 18 GPa.[3] This flexibility helps in supporting the more brittle, nonresilient enamel.
IV. The ultimate *tensile strength* of dentin is approximately 98 MPa, whereas the ultimate tensile strength of enamel is approximately 10 MPa.
V. The *compressive strength* of dentin and enamel are approximately 297 and 384 MPa, respectively.[3]

Dentin versus enamel

During tooth preparation, dentin usually is distinguished from enamel by

I. ***Colour and opacity***. Dentin is normally *yellow-white* and slightly darker than enamel. In older patients, dentin is darker, and it can become brown or black when it has been exposed to oral fluids, old restorative materials or slowly advancing caries.
II. ***Reflectance***. Dentin surfaces are more opaque and dull, being less reflective to light than similar enamel surfaces, which appear shiny.
III. ***Hardness***. Dentin is softer than enamel and provides greater yield to the pressure of a sharp explorer tine, which tends to catch and hold in dentin.
IV. ***Sound***. Enamel would yield a scratchy sound on probing with a sharp explorer tine, which is not apparent in dentin.

Hydrodynamic theory of dentinal sensitivity

Sensitivity is encountered whenever odontoblasts and their processes are stimulated during operative procedures, even though the pain receptor mechanism appears to be within the dentinal tubules near the pulp. Physical, thermal, chemical, bacterial and traumatic stimuli are transmitted through the dentinal tubules.

The most accepted theory of pain transmission is the *hydrodynamic theory*, which accounts for pain transmission through rapid movements of fluid within the dentinal tubules.[4] Dentinal tubules are filled with *dentinal fluid*, a transudate of plasma. When enamel or cementum is removed during tooth preparation, the external seal of dentin is lost, allowing tubular fluid to move towards the cut surface (Fig. 1.20). Pulpal fluid has a slight positive pressure that forces fluid outward towards any breach in the external seal.

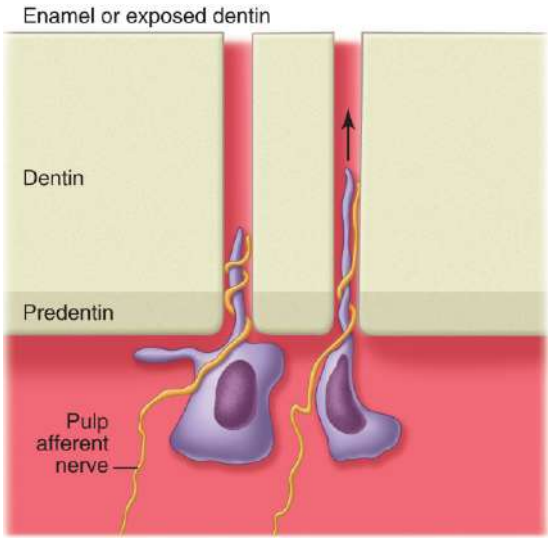


Figure 1.20 Stimuli that induce fluid movements in dentinal tubules distort odontoblasts and afferent nerves, leading to a sensation of pain. Many operative procedures such as cutting or air-drying induce such fluid movement *(arrow)*.

Dentinal permeability

The permeability of dentin is not uniform throughout the tooth.

I. Coronal dentin is much more permeable than root dentin. There are also differences within coronal dentin (Fig. 1.21).[5]
II. Dentin permeability primarily depends on the remaining dentin thickness (i.e. length of the tubules) and the diameter of the tubules.

As the tubules are shorter, more numerous and larger in diameter closer to the pulp, deep dentin is a less effective pulpal barrier than superficial dentin (Fig. 1.21).

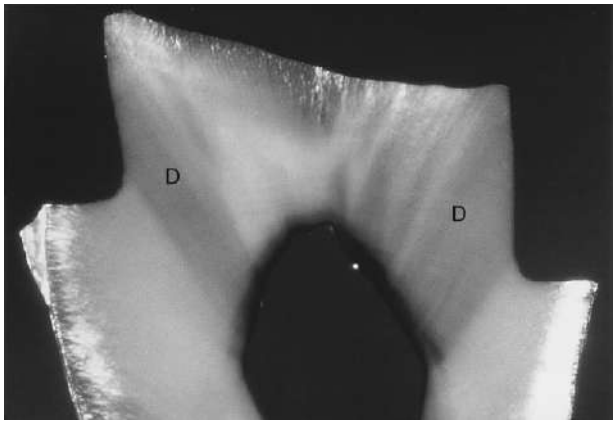


Figure 1.21 Ground section of MOD (mesio-occluso-distal) tooth preparation on the third molar. Dark blue dye was placed in the pulp chamber under pressure after tooth preparation. Dark areas of dye penetration (*D*) show that the dentinal tubules of axial walls are much more permeable than those of the pulpal floor of preparation.

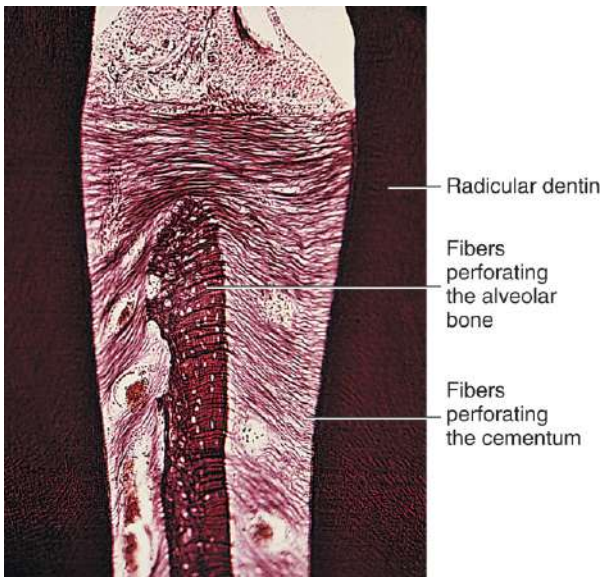


Figure 1.22 Principal fibres of periodontal ligament continue to course into surface layer of cementum as Sharpey's fibres.
(From Avery JK, Chiego DJ: *Essentials of oral histology and embryology: a clinical approach*, ed 3, St Louis, MO, 2006, Mosby.)

Cementum

Cementum is a thin layer of hard dental tissue covering the anatomic roots of teeth and is formed by cells known as *cementoblasts*, which develop from undifferentiated mesenchymal cells in the connective tissue of the dental follicle. Cementum is light yellow and slightly lighter in colour than dentin. *Sharpey's fibres* are portions of the principal collagenous fibres of the periodontal ligament embedded in cementum and alveolar bone to attach the tooth to the alveolus (Fig. 1.22).

Chemical composition

Cementum is slightly softer than dentin. It consists of the following:

- About 45%–50% inorganic material (hydroxyapatite) by weight.
- About 50%–55% organic matter and water by weight. The organic portion is composed primarily of collagen and protein polysaccharides.

Types of cementum

Cementum is avascular and is formed continuously throughout life because as the superficial layer of cementum ages, a new layer of cementum is deposited to keep the attachment intact. Two kinds of cementum are formed:

- ***Acellular cementum.*** The acellular layer of cementum does not incorporate cells into its structure and usually predominates on the coronal half of the root.
- ***Cellular cementum.*** Cellular cementum incorporates cells into its structure and occurs more frequently on the apical half of the root.

Clinical Notes

- Cementum thickness can increase on the root end to compensate for attritional wear of the occlusal or incisal surface and passive eruption of the tooth.
- Cementum joins enamel to form the CEJ, which is referred to as the *cervical line*. In about 10% of teeth, enamel and cementum do not meet, and this can result in a sensitive area as the openings of the dentinal tubules are not covered.
- Abrasion, erosion, caries, scaling and restoration finishing and polishing procedures can denude dentin of its cementum covering, which can cause the dentin to be sensitive to various stimuli (e.g. heat, cold, sweet substances, sour substances).

Physiology of Tooth Form

Function

Teeth serve four main functions:

I. ***Mastication.*** Normal tooth form and proper alignment ensure efficiency in the incising and reduction of food with the various tooth classes—incisors, canines, premolars and molars—performing specific functions in the masticatory process and in the coordination of the various muscles of mastication.
II. ***Esthetics.*** The form and alignment of the anterior teeth are important to a person's physical appearance.
III. ***Speech.*** The form and alignment of anterior and posterior teeth assist in the articulation of certain sounds that can have a significant effect on speech.
IV. ***Protection of supporting tissues.*** The form and alignment of the teeth assist in sustaining them in the dental arches by assisting in the development and protection of gingival tissue and alveolar bone that support them.

Contours

Facial and lingual surfaces possess a degree of convexity that affords protection and stimulation of supporting tissues during mastication. The convexity generally is located at

I. the cervical third of the crown on the facial surfaces of all teeth and
II. the lingual surfaces of incisors and canines.

Lingual surfaces of posterior teeth usually have their height of contour in the middle third of the crown. Normal tooth contours act in deflecting food only to the extent that the passing food stimulates (by gentle massage) and does not irritate (abrade) supporting soft tissues. If these curvatures are too great, tissues usually receive inadequate stimulation by the passage of food. Too little contour may result in trauma to the attachment apparatus. Normal tooth contours must be recreated in the performance of operative dental procedures. Improper location and degree of facial or lingual convexities may result in iatrogenic injury, as illustrated in Fig. 1.23, in which the proper facial contour is disregarded in the design of the cervical area of a mandibular molar restoration. Overcontouring is the worst offender, usually resulting in increased plaque retention that leads to a chronic inflammatory state of the gingiva.

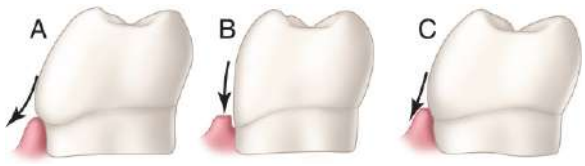


Figure 1.23 Contours. *Arrows* show pathways of food passing over facial surface of mandibular molar during mastication. (A) Overcontour deflects food from gingiva and results in understimulation of supporting tissues. (B) Undercontour of tooth may result in irritation of soft tissue. (C) Correct contour permits adequate stimulation and protection of supporting tissue.

Proper form of the proximal surfaces of teeth is just as important to the maintenance of periodontal tissue health as is the proper form of facial and lingual surfaces. The proximal height of contour serves to provide

I. contacts with the proximal surfaces of adjacent teeth, thus preventing food impaction;
II. adequate embrasure space (immediately apical to the contacts) for gingival tissue, supporting bone, blood vessels and nerves that serve the supporting structures (Fig. 1.24).

Proximal Contact Area

When teeth initially erupt to make proximal contact with previously erupted teeth, a contact *point* is present. The contact point increases in size to become a proximal contact *area* as the two adjacent tooth surfaces abrade each other during physiologic tooth movement (Figs. 1.25 and 1.26).



Figure 1.24 Portion of the skull, showing triangular spaces beneath proximal contact areas. These spaces are occupied by soft tissue and bone for the support of teeth.

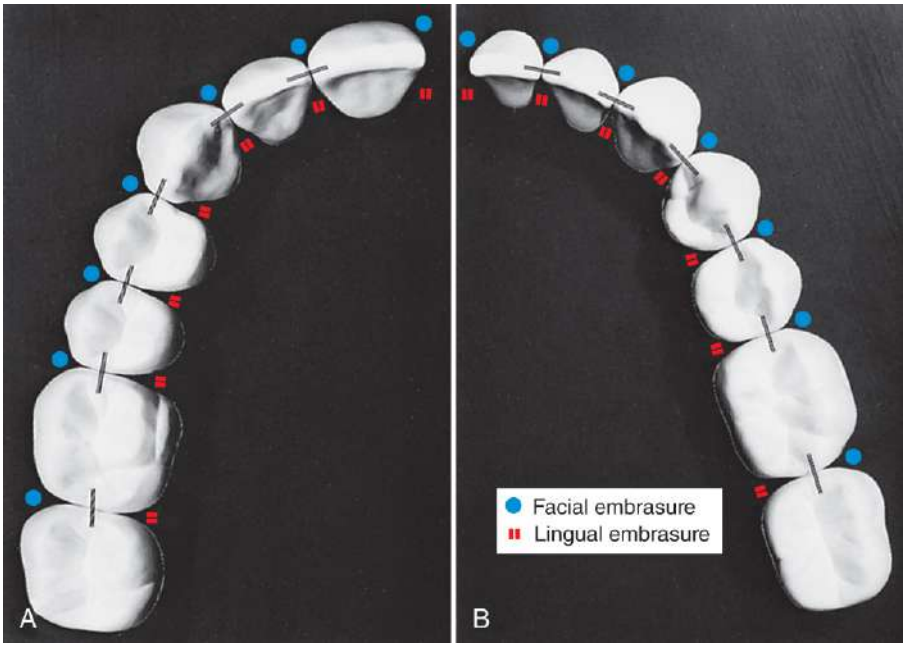


Figure 1.25 Proximal contact areas. *Black lines* show positions of contact faciolingually. (A) Maxillary teeth. (B) Mandibular teeth. Facial and lingual embrasures are indicated.

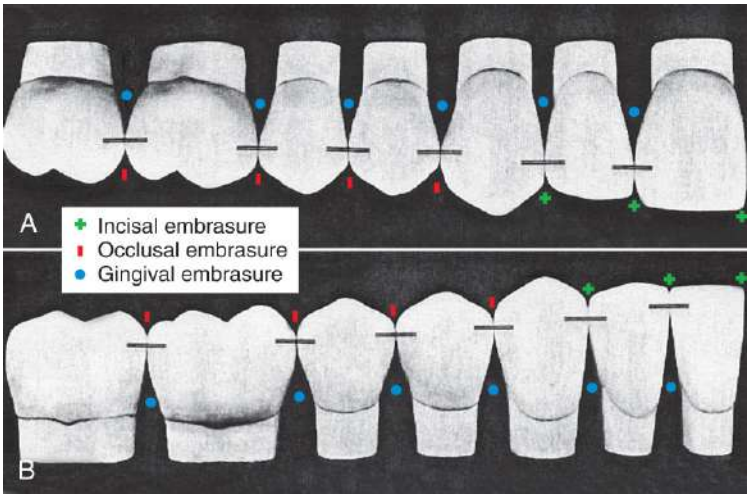


Figure 1.26 Proximal contact areas. *Black lines* show positions of contact incisogingivally and occlusogingivally. Incisal, occlusal and gingival embrasures are indicated. (A) Maxillary teeth. (B) Mandibular teeth.

The physiologic significance of properly formed and located proximal contacts cannot be overemphasized; they promote normal healthy interdental papillae filling the interproximal spaces. Improper contacts may result in food impaction between teeth, potentially increasing the risk of periodontal disease, caries and tooth movement. In addition, retention of food is objectionable because of its physical presence and the halitosis that results from food decomposition. Proximal contacts and interdigitation of maxillary and mandibular teeth, through occlusal contact areas, stabilize and maintain the integrity of the dental arches.

The proximal contact area is located in the following regions:

I. *Incisal third* of the approximating surfaces of maxillary and mandibular central incisors (Fig. 1.26). It is positioned slightly facial to the centre of the proximal surface faciolingually (Fig. 1.25).
II. Proceeding posteriorly from the incisor region through all the remaining teeth, the contact area is located near the *junction of the incisal (or occlusal) and middle thirds* or in the *middle third*.
III. Proximal contact areas typically are larger in the molar region, which helps prevent gingival food impaction during mastication. Adjacent surfaces near the proximal contacts (embrasures) usually have remarkable symmetry.

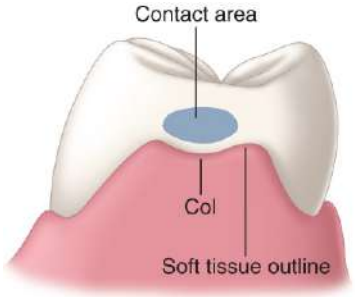


Figure 1.27 Relationship of ideal interdental papilla to molar contact area.

Embrasures

Embrasures are V-shaped spaces that originate at the proximal contact areas between adjacent teeth and are named for the direction towards which they radiate. These embrasures are

I. facial,
II. lingual,
III. incisal or occlusal and
IV. gingival (Figs. 1.25 and 1.26).

Initially, the interdental papilla fills the gingival embrasure. When the form and function of teeth are ideal and optimal oral health is maintained, the interdental papilla may continue in this position throughout life. When the gingival embrasure is filled by the papilla, trapping of food in this region is prevented. In a faciolingual vertical section, the papilla is seen to have a triangular shape between anterior teeth, whereas in posterior teeth, the papilla may be shaped like a mountain range, with facial and lingual peaks and the *col* (valley) lying beneath the contact area (Fig. 1.27).

This col, a central faciolingual concave area beneath the contact, is more vulnerable to periodontal disease from incorrect contact and embrasure form because it is covered by non-keratinized epithelium.

The correct relationships of embrasures, cusps to sulci, marginal ridges and grooves of adjacent and opposing teeth provide for the escape of food from the occlusal surfaces during mastication. When an embrasure is decreased in size or absent, additional stress is created on teeth and the supporting structures during mastication. Embrasures that are too large provide little protection to the supporting structures as food is forced into the interproximal space by an opposing cusp (Fig. 1.28). A prime example is the failure to restore the distal cusp of a mandibular first molar when placing a restoration (Fig. 1.29). Lingual embrasures are usually larger than facial embrasures; and this allows more food to be displaced lingually because the tongue can return the food to the occlusal surface more easily than if the food is displaced facially into the buccal vestibule (Fig. 1.25). The marginal ridges of adjacent posterior teeth should be at the same height to have proper contact and embrasure forms. When this relationship is absent, it may

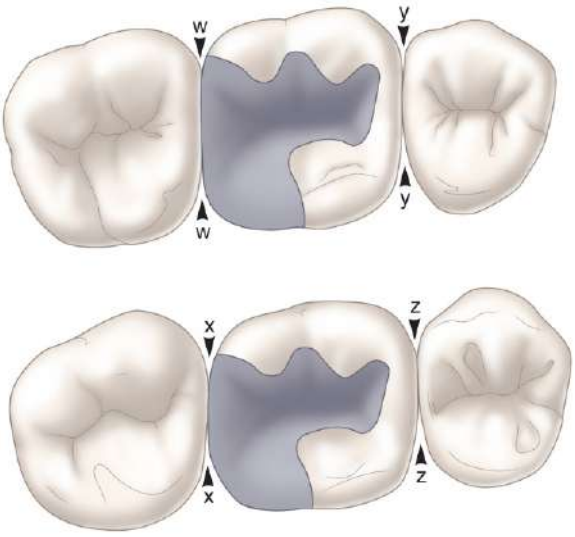


Figure 1.28 Embrasure form. *w*, improper embrasure form caused by overcontouring of restoration resulting in unhealthy gingiva from lack of stimulation; *x*, good embrasure form; *y*, frictional wear of contact area has resulted in decrease of embrasure dimension; *z*, when the embrasure form is good, supporting tissues receive adequate stimulation from foods during mastication.

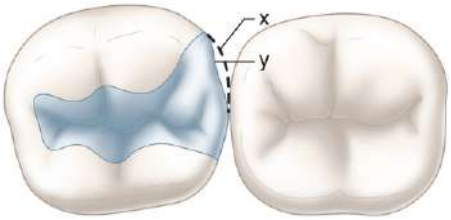


Figure 1.29 Embrasure form. *x,* Portion of tooth that offers protection to underlying supporting tissue during mastication; *y,* restoration fails to establish adequate contour for good embrasure form.

cause an increase in the problems associated with inadequate proximal contacts and faulty embrasure forms.

Preservation of the curvatures of opposing cusps and surfaces in function maintains masticatory efficiency throughout life. Correct anatomic form renders teeth more self-cleansing because of the smoothly rounded contours that are more exposed to the cleansing action of foods and fluids and the frictional movement of the tongue, lips and cheeks. Failure to understand and adhere to correct anatomic form may contribute to the breakdown of the restored system (Fig. 1.30).

Periodontium

The periodontium consists of the oral hard and soft tissues that invest and support teeth. It can be divided into the following:

I. The *gingival unit*, consisting of
   i. free gingiva,
   ii. attached gingiva and
   iii. alveolar mucosa.

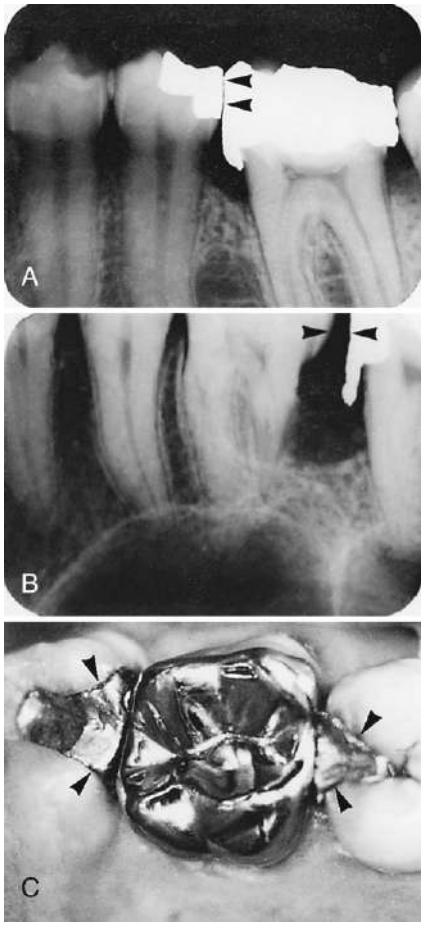


Figure 1.30 Poor anatomic restorative form. (A) Radiograph of flat contact/amalgam gingival excess and resultant vertical osseous loss. (B) Radiograph of restoration with amalgam gingival excess and absence of contact resulting in osseous loss, adjacent root caries. (C) Poor embrasure form and restoration margins.

II. The *attachment apparatus*, consisting of
   i. cementum,
   ii. periodontal ligament and
   iii. alveolar process (Fig. 1.31).

Gingival Unit

Free gingiva

I. The free gingiva is the gingiva from the marginal crest to the level of the base of the gingival sulcus (Fig. 1.31, *labels 4 and 6*).
II. The *gingival sulcus* is the space between the tooth and the free gingiva.
III. The outer aspect of the free gingiva in each gingival embrasure is called *gingival* or *interdental papilla*.
IV. The *free gingival groove* is a shallow groove that runs parallel to the marginal crest of the free gingiva and usually indicates the level of the base of the gingival sulcus (Fig. 1.31, *label 7*).

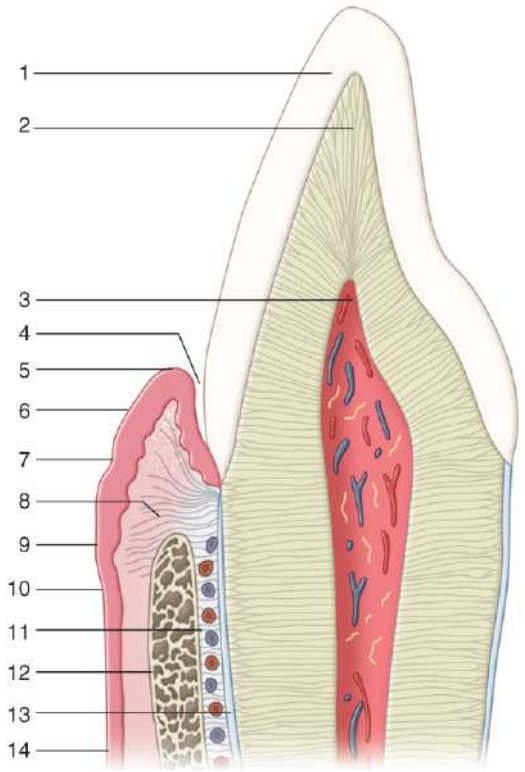


Figure 1.31 Vertical section of a maxillary incisor illustrating supporting structures: 1, Enamel; 2, dentin; 3, pulp; 4, gingival sulcus; 5, free gingival margin; 6, free gingiva; 7, free gingival groove; 8, lamina propria of gingiva; 9, attached gingiva; 10, mucogingival junction; 11, periodontal ligament; 12, alveolar bone; 13, cementum and 14, alveolar mucosa.

Attached gingiva

The attached gingiva, a dense connective tissue with keratinized, stratified, squamous epithelium, extends from the depth of the gingival sulcus to the mucogingival junction. A dense network of collagenous fibres connects the attached gingiva firmly to cementum and the periosteum of the alveolar process (bone).

Alveolar mucosa

The alveolar mucosa is a thin, soft tissue that is loosely attached to the underlying alveolar bone (Fig. 1.31, *labels 12 and 14*). It is covered by a thin, non-keratinized epithelial layer. The alveolar mucosa is delineated from the attached gingiva by the mucogingival junction and continues apically to the vestibular fornix and the inside of the cheek.

Clinical Notes

The level of the gingival attachment and gingival sulcus is an important factor in restorative dentistry. The margin of a tooth preparation **should not be positioned subgingivally** (at levels between the marginal crest of the free gingiva and the base of the sulcus) unless dictated by caries, previous restoration, esthetics or other preparation requirements.

Attachment Apparatus

Periodontal ligament

The tooth root is attached to the alveolus (bony socket) by the periodontal ligament (Fig. 1.31, *label 11*), which is a complex connective tissue containing numerous cells, blood vessels, nerves and an extracellular substance consisting of fibres and ground substance. Most of the fibres are collagen, and the ground substance is composed of a variety of proteins and polysaccharides.

The periodontal ligament serves the following functions:

I. ***Attachment and support.*** Bundles of collagen fibres, known as *principal fibres of the ligament*, serve to attach cementum to alveolar bone and act as a cushion to suspend and support the tooth.
II. ***Sensory.*** Bundles of collagen fibres, known as *principal fibres of the ligament*, serve to attach cementum to alveolar bone and act as a cushion to suspend and support the tooth.
III. ***Nutritive.*** Blood vessels supply the attachment apparatus with nutritive substances.
IV. ***Homeostatic.*** Specialized cells of the ligament function to resorb and replace cementum, the periodontal ligament and alveolar bone.

Alveolar process

The alveolar process—a part of the maxilla and the mandible—forms, supports and lines the sockets into which the roots of teeth fit. The alveolar process comprises thin, compact bone with many small openings through which blood vessels, lymphatics and nerves pass. It consists of two parts:

I. ***Alveolar bone proper.*** This is the inner wall of the bony socket which consists of the thin lamella of bone that surrounds the root of the tooth.
II. ***Supporting alveolar bone.*** This is the part of the alveolar process which surrounds the alveolar bone proper and supports the socket. Supporting bone is composed of two parts:
    i. *Cortical plate.* This consists of compact bone and forms the inner (lingual) and outer (facial) plates of the alveolar process.
    ii. *Spongy base.* This fills the area between the plates and the alveolar bone proper.

Occlusion

Occlusion literally means *closing;* in dentistry, the term means the contact of teeth in opposing dental arches when the jaws are closed (static occlusal relationships) and during various jaw movements (dynamic occlusal relationships).

A wide variety of occlusal schemes can be found in healthy individuals. Failing to find a single adequate definition of an ideal occlusal scheme has resulted in the conclusion that *in the final analysis, optimal function and the absence of disease is the principal characteristic of a good occlusion.*[6]

Tooth Alignment and Dental Arches

Central Groove (Fig. 1.32A)

The cusps in the posterior teeth are separated by distinct developmental grooves and sometimes have additional supplemental grooves on the cusp inclines. The facial cusps are separated from the lingual cusps by a deep groove, termed *central groove*.

Fossae

If a tooth has multiple facial cusps or multiple lingual cusps, the cusps are separated by facial or lingual developmental grooves. The depressions between the cusps are termed *fossae* (singular, is *fossa*).

Facial Occlusal Line

The maxillary arch is larger than the mandibular arch, which results in the maxillary cusps overlapping the mandibular cusps when the arches are in maximal occlusal contact (Fig. 1.32B). In Fig. 1.32A, two curved lines have been drawn over the teeth to aid in the visualization of the arch form. These curved lines identify the alignment of similarly functioning cusps or fossae. On the left side of the arches, an imaginary arc connecting the row of facial cusps in the mandibular arch have been drawn and labelled *facial occlusal line*.

Central Fossa Occlusal Line

An imaginary line connecting the maxillary central fossae is labelled *central fossa occlusal line*. *The mandibular facial occlusal line and the maxillary central fossa occlusal line coincide exactly when the mandibular arch is fully closed into the maxillary arch.* On the right side of the dental arches, the maxillary lingual occlusal line and mandibular central fossa occlusal line have been drawn and labelled. These lines also coincide when the mandible is fully closed.

Maximum Inter-cuspation

This refers to the position of the mandible when teeth are brought into full interdigitation with the maximal number of teeth contacting. Synonyms for maximum inter-cuspation (MI) include *inter-cuspal contact*, *maximum closure* and *maximum habitual inter-cuspation (MHI)*.

I. ***Functional cusps.*** Cusps that contact opposing teeth along the central fossa occlusal line are termed *functional cusps* (supporting, centric, holding or stamp cusps).
II. ***Nonfunctional cusps.*** Cusps that overlap opposing teeth are termed *nonfunctional cusps* (nonsupporting, noncentric or non-holding cusps).

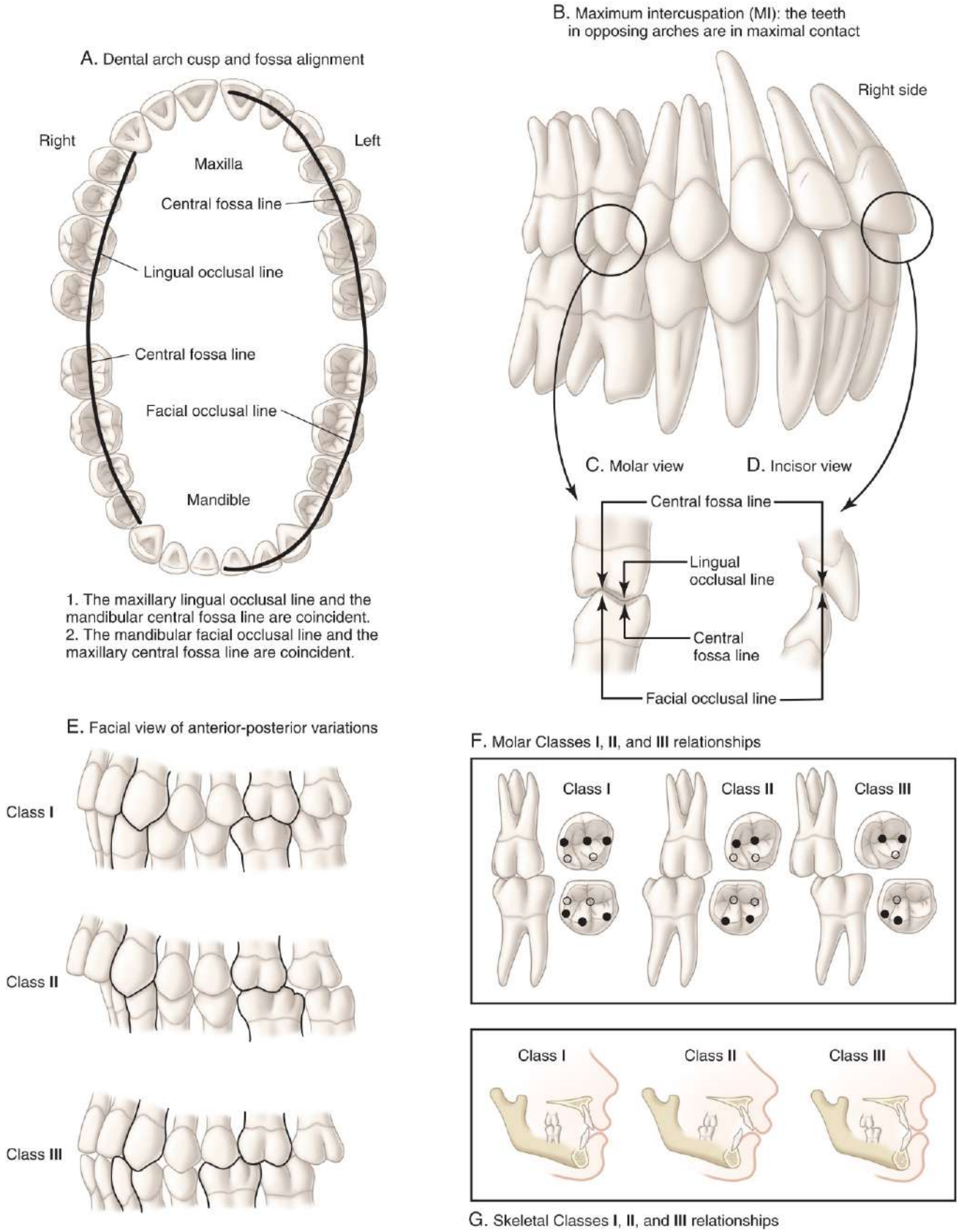


Figure 1.32 Dental arch relationships. (A) Dental arch cusp and fossa alignment; (B) maximum inter-cuspation; (C) molar view; (D) incisor view; (E) facial view of anteroposterior variations; (F) molar classes I, II and III relationships and (G) skeletal classes I, II and III relationships.

The mandibular facial occlusal line identifies the mandibular functional cusps, whereas the maxillary facial cusps are nonfunctional cusps. These terms are usually applied only to posterior teeth to distinguish the functions of the two rows of cusps.

Anteroposterior Interarch Relationships

The cusp interdigitation pattern of the first molar teeth is used to classify anteroposterior arch relationships using a system developed by Angle.[7] The location of the mesiofacial cusp of the maxillary first molar in relation to the mandibular first molar is used as an indicator in Angle's classification. Three interdigitated relationships of the first molars are commonly observed (Fig. 1.32E and F).

- **Angle class I.** The most common molar relationship finds the maxillary mesiofacial cusp located in the mesiofacial developmental groove of the mandibular first molar. This relationship is termed Angle class I.
- **Angle class II.** Slight posterior positioning of the mandibular first molar results in the mesiofacial cusp of the maxillary molar settling into the facial embrasure between the mandibular first molar and the mandibular second premolar. This is termed Angle class II.
- **Angle class III.** Anterior positioning of the mandibular first molar relative to the maxillary first molar is termed as Angle class III and is the least common. In class III relationships, the mesiofacial cusp of the maxillary first molar fits into the distofacial groove of the mandibular first molar.

Although Angle's classification is based on the relationship of the cusps. Fig. 1.24G illustrates that the location of tooth roots in alveolar bone determines the relative positions of the crowns and cusps of teeth:

- **Class I molar relationship.** When the mandible is proportionally similar in size to the maxilla, a class I molar relationship is formed.
- **Class II molar relationship.** *When the mandible is proportionally smaller than the maxilla, a class II molar relationship is formed.*
- **Class III molar relationship.** When the mandible is relatively greater than the maxilla, a class III molar relationship is formed.

Interarch Tooth Relationships

Fig. 1.33 illustrates the occlusal contact relationships of individual teeth in more detail. In Fig. 1.33, A-2, incisor overlap is illustrated. The overlap is characterized in two dimensions:

I. Horizontal overlap (*overjet*)
II. Vertical overlap (*overbite*)

Differences in the sizes of the mandible and the maxilla can result in clinically significant variations in incisor relationships, including *open bite* as a result of mandibular deficiency or excessive eruption of posterior teeth, and *crossbite* as a result of mandibular growth excess (Fig. 1.33, A-3). These variations have significant clinical effects on the contacting relationships of posterior teeth during various jaw movements because anterior teeth do not provide gliding contact. The various variations in premolar and molar relationships are illustrated in Figs. 1.33B and C.

Posterior Cusp Characteristics

Four cusp ridges can be identified as common features of all the cusps:

Outer Incline Ridge

The outer incline of a cusp faces the facial (or the lingual) surface of the tooth and is named for its respective surface. In the example using a mandibular second premolar (Fig. 1.34A), the facial cusp ridge of the facial cusp is indicated by the line that points to the outer incline of the cusp.

Inner Incline Ridge

The inner inclines of the posterior cusps face the central fossa or the central groove of the tooth. The inner incline cusp ridges are widest at the base and become narrower as they approach the cusp tip. For this reason, they are also termed as *triangular ridges*. Triangular ridges are usually set off from the other cusp ridges by one or more supplemental grooves. In Fig. 1.34, *B-1* and *C-1*, the outer inclines of the facial cusps of the mandibular and maxillary first molars are highlighted. In Fig. 1.34, *B-2* and *C-2*, the triangular ridges of the facial and lingual cusps are highlighted.

Mesial Cusp Ridges and Distal Cusp Ridges

The mesial and distal cusp ridges extend from the cusp tip mesially and distally and are named for their directions. The mesial and distal cusp ridges extend downward from the cusp tips, forming the characteristic facial and lingual profiles of the cusps as viewed from the facial or lingual aspect. At the base of the cusp, the mesial or distal cusp ridge abuts to another cusp ridge, forming a developmental groove, or the cusp ridge turns towards the centre line of the tooth and fuses with the marginal ridge.

Marginal ridges are elevated; the rounded ridges being located on the mesial and distal edges of the tooth's occlusal surface (Fig. 1.34A).

Functional Cusps

In Fig. 1.35, the lingual occlusal line of maxillary teeth and the facial occlusal line of mandibular teeth mark the locations of the functional cusps. These cusps contact opposing teeth in their corresponding faciolingual centre on a marginal

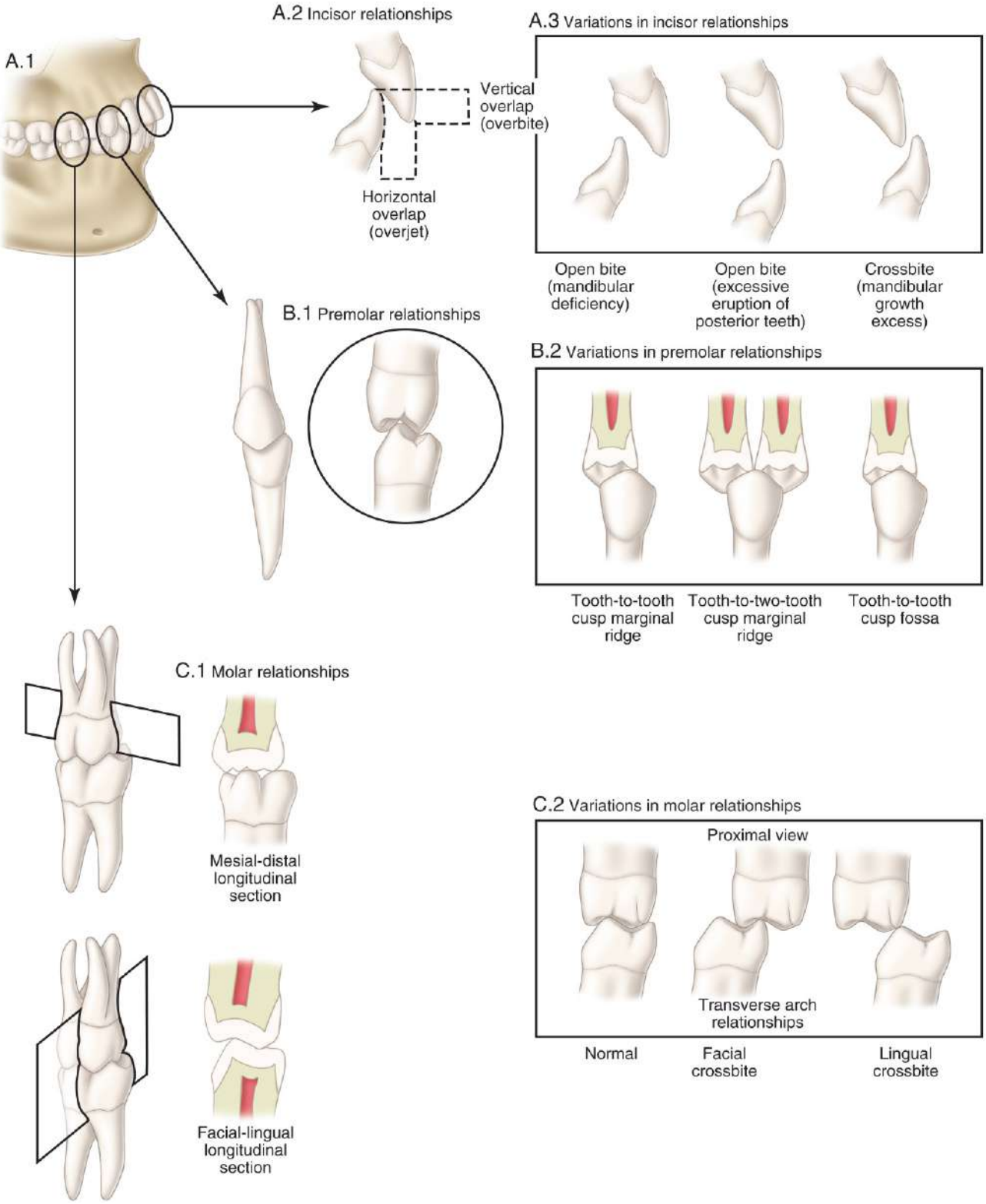


Figure 1.33 Tooth relationships.

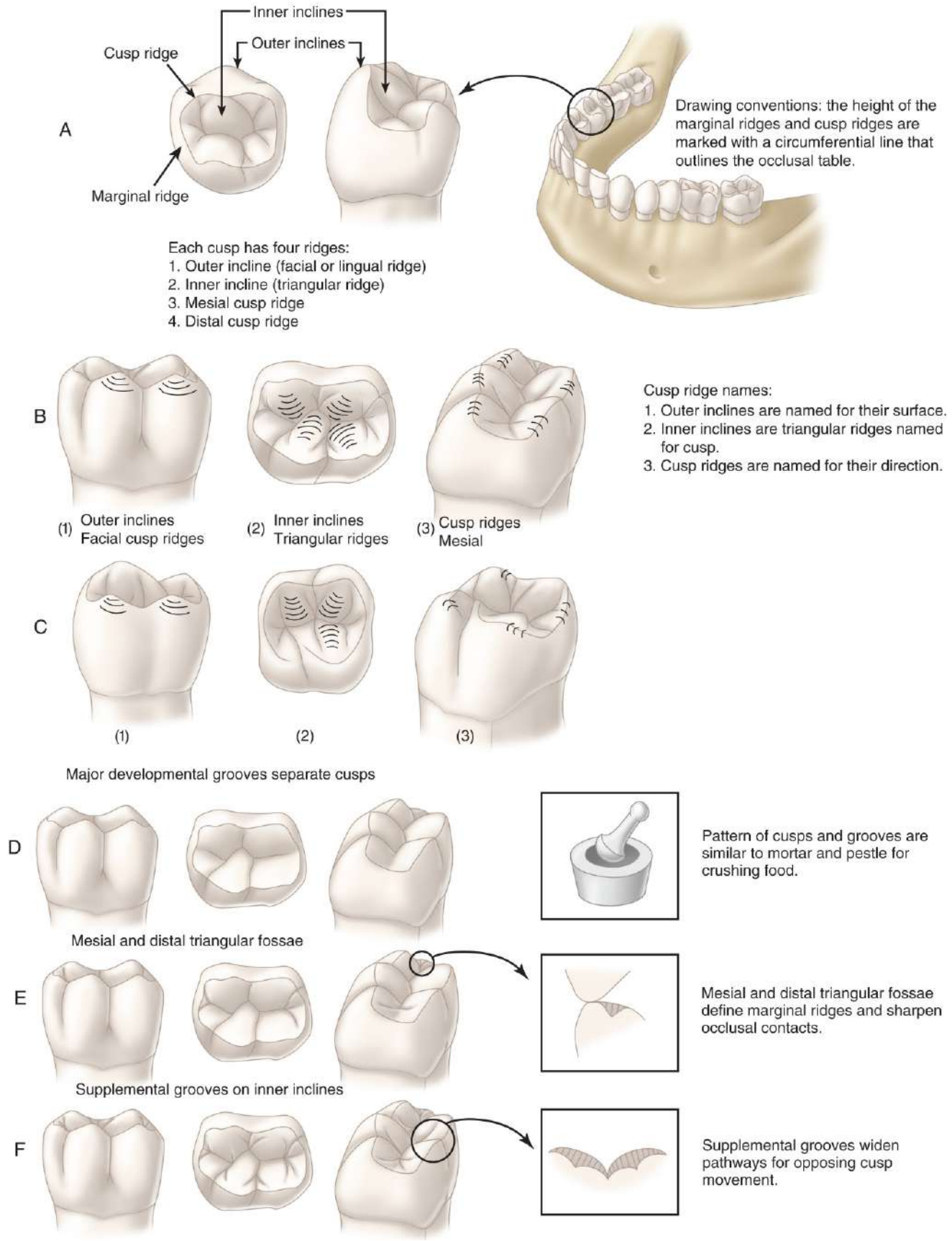


Figure 1.34 Common features of all posterior teeth.

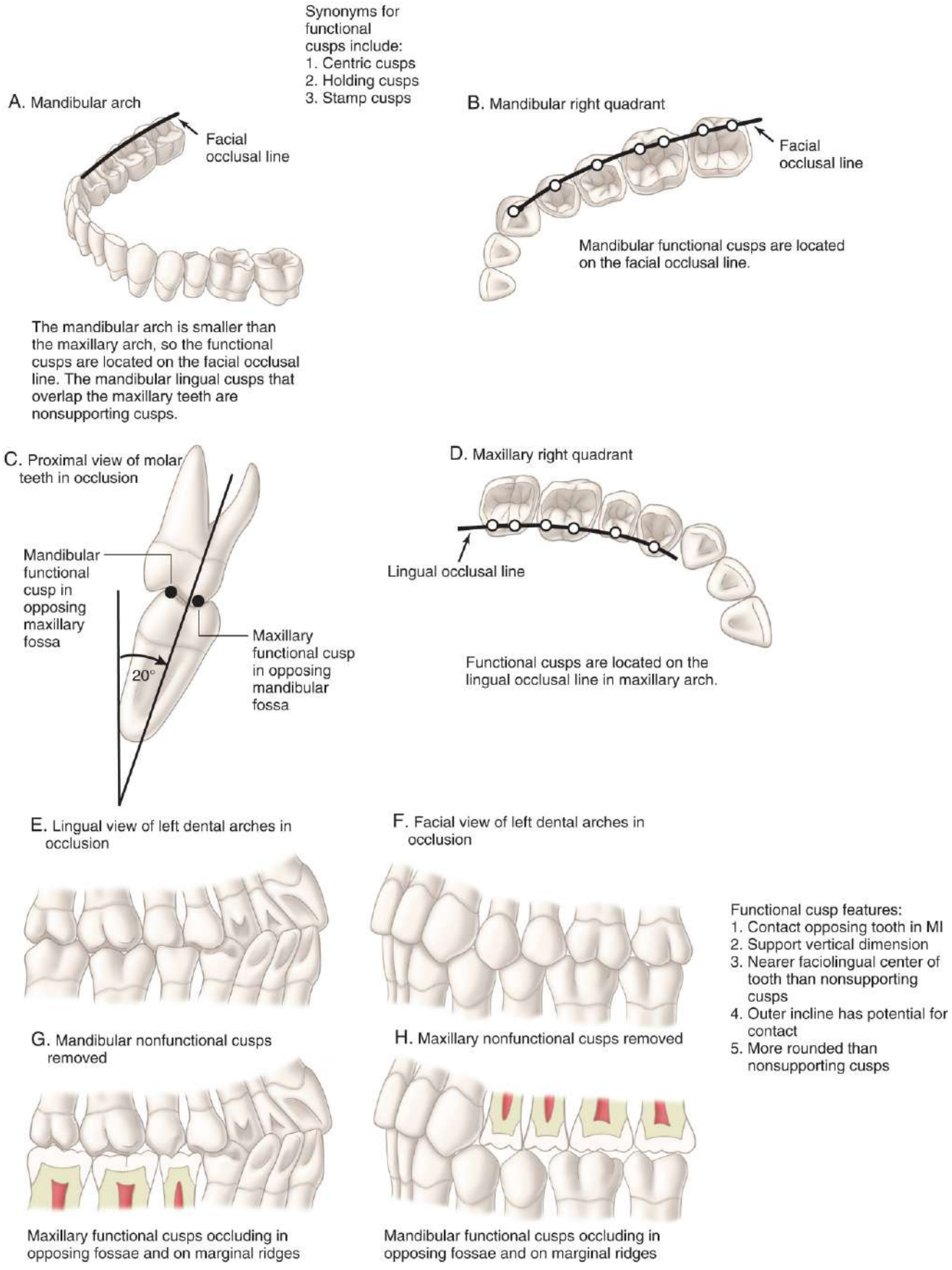


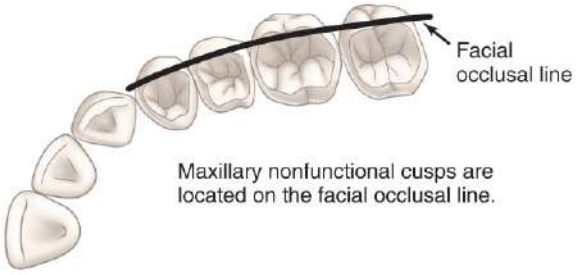
Figure 1.35 Functional cusps.

A. Maxillary arch

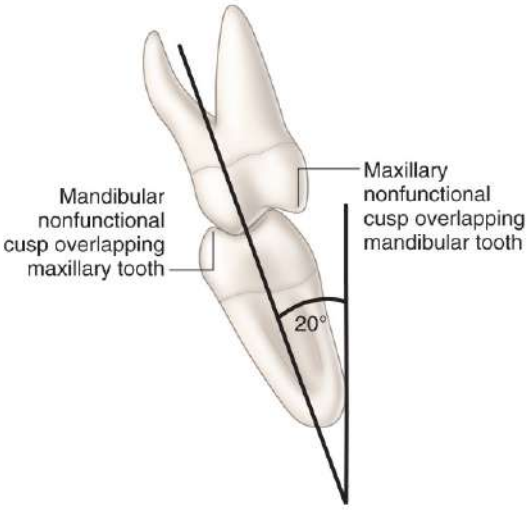


The maxillary arch is larger than the mandibular arch causing the maxillary facial line (nonfunctional cusps) to overlap the mandibular teeth.

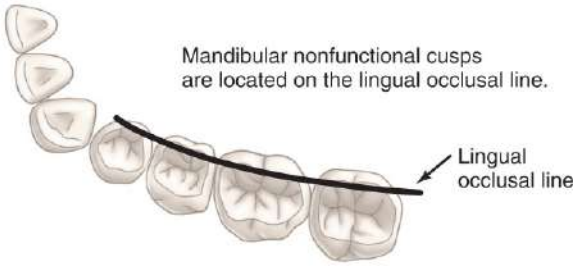
B. Maxillary left quadrant



C. Molar teeth in occlusion

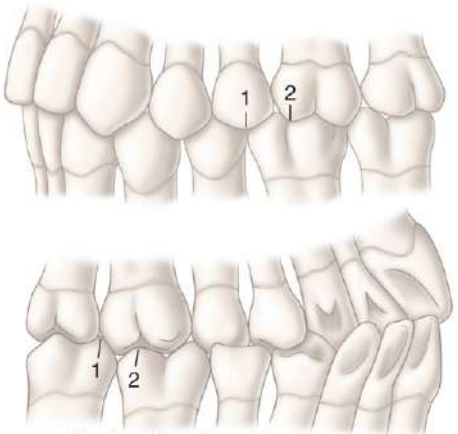


D. Mandibular left quadrant



Nonfunctional cusp features:
1. Do not contact opposing tooth in MI
2. Keep soft tissue of tongue or cheek off occlusal table
3. Farther from faciolingual center of tooth than supporting cusps
4. Outer incline has no potential for contact
5. Have sharper cusp ridges than supporting cusps

E. Views of left dental arches in occlusion showing interdigitation of nonfunctional cusps



Nonfunctional cusp location:
1. Opposing embrasure
2. Opposing developmental groove

F. Views of left dental arches in occlusion showing facial and lingual occlusal lines

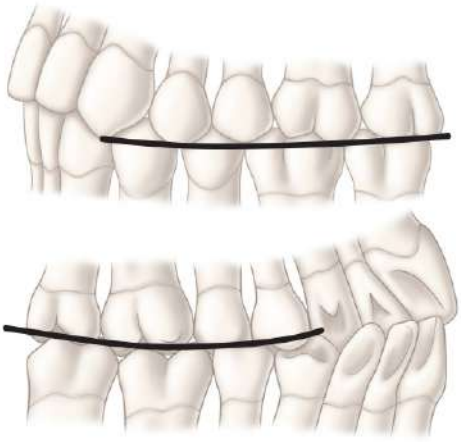


Figure 1.36 Nonfunctional cusps.

ridge or a fossa. Functional cusp–central fossa contact has been compared to a mortar and pestle because the supporting cusp cuts, crushes and grinds fibrous food against the ridges forming the concavity of the fossa (Fig. 1.34D). Functional cusps also serve to prevent drifting and passive eruption of teeth, hence the term *holding cusps*. Functional cusps (Fig. 1.35) can be identified by five characteristic features[8]:

I. They contact the opposing tooth in MI.
II. They support the vertical dimension of the face.
III. They are nearer the faciolingual centre of the tooth than nonsupporting cusps.
IV. Their outer incline has the potential for contact.
V. They have broader, more rounded cusp ridges than nonsupporting cusps.

Clinical Notes

- As the maxillary arch is larger than the mandibular arch, the maxillary functional cusps are located on the maxillary lingual occlusal line (Fig. 1.35D).
- The mandibular functional cusps are located on the mandibular facial occlusal line (Fig. 1.35A and B).
- The lingual tilt of posterior teeth increases the relative height of the functional cusps with respect to the nonfunctional cusps (Fig. 1.35C), and the central fossa contacts of the functional cusps are obscured by the overlapping nonfunctional cusps.

Nonfunctional Cusps

The nonfunctional cusps overlap the opposing tooth without contacting the tooth. Fig. 1.36 illustrates that the nonfunctional cusps form a lingual occlusal line in the mandibular arch (Fig. 1.36D) and a facial occlusal line in the maxillary arch (Fig. 1.36B). The nonfunctional cusps are located in the anteroposterior plane in facial (lingual) embrasures or in the developmental groove of opposing teeth, creating an alternating arrangement when teeth are in MI.

Summary

An intimate knowledge of the anatomy and physiology of the dental tissues is important for a clinician to be able to use various dental materials optimally. A thorough understanding of the tooth anatomy and their occlusal relationships is crucial for a restorative dentist in being able to perform adequate restorative rehabilitation.

References

1. Garberoglio R, Brännström M: Scanning electron microscopic investigation of human dentinal tubules, *Arch Oral Biol* 21:355–362, 1976.
2. Scott JH, Symons NBB: *Introduction to dental anatomy*, ed 9, Philadelphia, PA, 1982, Churchill Livingstone.
3. Craig RG, Powers JM: *Restorative dental materials*, ed 12, St Louis, MO, 2006, Mosby.
4. Brännström M: *Dentin and pulp in restorative dentistry*, London, 1982, Wolfe Medical.
5. Sturdevant JR, Pashley DH: Regional dentin permeability of class I and II cavity preparations (abstract no. 173), *J Dent Res* 68:203, 1989.
6. Mohl ND, Zarb GA, Carlsson GE, et al: The dentition. In Mohl ND, Zarb GA, Carisson GE, et al, editors: *A textbook of occlusion*, Chicago, IL, 1988, Quintessence.
7. Angle EH: Classification of malocclusion, *Dent Cosmos* 41, 1899, 248–264, 350–357.
8. Kraus BS, Jorden E, Abrams L: *Dental anatomy and occlusion*, ed 1, Baltimore, MD, 1969, Williams & Wilkins.
9. Digka A, Lyroudia K, Jirasek T, et al: Visualisation of human dental pulp vasculature by immunohistochemical and immunofluorescent detection of CD34: a comparative study, *Aust Endod J* 32:101–106, 2006.

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